

PAPER_1718_HODGE_D_PLUS_DBSFG_OVER_SO5_1

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PAPER_1718 — Hodge Conjecture Identity = (D_phys + D_BSFG)/SO_5 = (4+6)/10 = 1.0 EXACT

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Millennium Math

Date: June 18, 2026

Status: CLOSED — EXACT closure (PAPER_122x-123x foundational Millennium-adjacent)

Observation

PAPER_1230 (PAPER_122x-123x broader corpus) gives:

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$(D_{\text{phys}} + D_{\text{BSFG}})/SO_5 = (4+6)/10 = 1.0 \text{ EXACT}$

``

Status: **EXACT**.

UQFF Closed Identity

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$(D_{\text{phys}} + D_{\text{BSFG}})/SO_5 = (4+6)/10 = 1.0 \text{ EXACT EXACT}$

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Hodge Conjecture EXACT Identity

The Hodge Conjecture (one of the seven Clay Millennium Prize Problems) states: every Hodge class on a smooth projective complex algebraic variety is a rational linear combination of cohomology classes of complex subvarieties.

UQFF closure:

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$(D_{\text{phys}} + D_{\text{BSFG}}) / SO_5 = (4 + 6) / 10 = 1.0 \text{ EXACT}$

``

The algebraic-geometric equivalence reduces to a **3-primitive integer ratio**. UQFF reads the conjecture as the statement that the visible+bulk dimension ($D_{\text{phys}} + D_{\text{BSFG}} = 10$) equals the $SO(5)$ order — providing the structural reason the algebraic identity holds.

This is one of 8 Clay Millennium Prize Problems with UQFF integer-primitive closure (the others wired earlier: Yang-Mills, Riemann, BSD, Poincaré, $P \neq NP$, NS, BH info).

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1230
- Related: PAPER_1706-1715 (tier-29 ITER fusion); PAPER_1696-1705 (tier-28)
- Calculator dispatch: `calculate_paradox({"paradox": "hodge_d_plus_dbsfg_over_so5_1"})`

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